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**Q1. String Character Frequency Counter**

CODE

```
s = input().lower()

count = {}

for ch in s:
    if ch != " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1

for ch in sorted(count):
    print(ch, "-", count[ch])
```

INPUT

Hello World

OUTPUT

```
d - 1
e - 1
h - 1
l - 3
o - 2
r - 1
w - 1
```

## Q2. Vowel and Consonant Classifier with Statistics

## CODE

```
s = input().lower()

vowels = "aeiou"
vowel_count = 0
consonant_count = 0
freq = {}

for ch in s:
    if ch.isalpha():
        freq[ch] = freq.get(ch, 0) + 1
        if ch in vowels:
            vowel_count += 1
        else:
            consonant_count += 1

# Most frequent vowel
max_vowel = "None"
max_count = 0
for v in vowels:
    if freq.get(v, 0) > max_count:
        max_count = freq[v]
        max_vowel = v

# Least frequent consonant
min_cons = "None"
min_count = float('inf')
for c in "bcdfghjklmnpqrstvwxyz":
    if freq.get(c, 0) > 0 and freq[c] < min_count:
        min_count = freq[c]
        min_cons = c

if consonant_count == 0:
    ratio = 0.00
else:
    ratio = round(vowel_count / consonant_count, 2)

print("Vowels:", vowel_count)
print("Consonants:", consonant_count)
print("Most frequent vowel:", max_vowel, "(", max_count, ")")
if min
```

## INPUT

Python Programming

## OUTPUT

```
File "C:\Windows\TEMP\sandbox_192524188RC_6a61e57de236b5.58828097\main.py", line 40
    if min
    ^
SyntaxError: expected ':'
```

**Q3. String Compression and Run-Length Encoding****CODE**

```
s = input().lower()

vowels = "aeiou"
v = 0
c = 0
vf = {}
cf = {}

for ch in s:
    if ch.isalpha():
        if ch in vowels:
            v += 1
            vf[ch] = vf.get(ch, 0) + 1
        else:
            c += 1
            cf[ch] = cf.get(ch, 0) + 1

print("Vowels:", v)
print("Consonants:", c)

if vf:
    mv = max(vf, key=vf.get)
    print("Most frequent vowel:", mv, "(", vf[mv], ")")
else:
    print("Most frequent vowel: None")

if cf:
    lc = min(cf, key=cf.get)
    print("Least frequent consonant:", lc, "(", cf[lc], ")")
else:
    print("Least frequent consonant: None")

if c == 0:
    ratio = 0.00
else:
    ratio = v / c

print("Vowel-to-Consonant Ratio:", format(ratio, ".2f"))
```

**INPUT**

aabbbccddddddee

**OUTPUT**

```
Vowels: 4
Consonants: 10
Most frequent vowel: a ( 2 )
Least frequent consonant: c ( 2 )
Vowel-to-Consonant Ratio: 0.40
```

## Q4. Advanced Text Statistics Analyser

## CODE

```

import re
from collections import Counter

text = input().strip()

paragraphs = 1 if text else 0
sentences = len(re.findall(r'[.!?]', text))
if sentences == 0 and text:
    sentences = 1

words = re.findall(r'[A-Za-z]+', text)
word_count = len(words)

total_letters = sum(len(w) for w in words)
avg_word = total_letters / word_count if word_count else 0

avg_sentence = word_count / sentences if sentences else 0

stop = {"a", "an", "in", "of", "and", "to", "the", "is", "it", "on", "for", "with", "at", "by"}

freq = Counter()
syllables = 0

for w in words:
    lw = w.lower()
    if lw not in stop:
        freq[lw] += 1

    c = 0
    prev = False
    for ch in lw:
        if ch in "aeiou":
            if not prev:
                c += 1
            prev = True
        else:
            prev = False
    if c == 0:
        c = 1
    syllables += c

top = freq.most_common(5)

score = 206.835 - 1.015 * (word_count / sentences) - 84.6 * (syllables / word_count)

print("Words:", word_count, "| Sentences:", sentences, "| Paragraphs:", paragraphs)
print("Avg Word Length:", round(avg_word, 2), "| Avg Sentence Length:", round(avg_sentence, 2))

print("Top 5 Words:", end=" ")
for i, (w, c) in enumerate(top):
    if i:
        print(", ", end=" ")
    print(f"{w}({c})", end="")
print()

print("Flesch Reading Ease:", round(score, 1))

```

## INPUT

Python is a powerful programming language.  
 Python supports data science.  
 Programming with Python is easy.  
 Python code is readable.

## OUTPUT

Words: 6 | Sentences: 1 | Paragraphs: 1  
 Avg Word Length: 6.0 | Avg Sentence Length: 6.0  
 Top 5 Words: python(1), powerful(1), programming(1), language(1)  
 Flesch Reading Ease: 31.5